
Additional Problems for Ch 10 HRW – Rotation
 Addisionele Probleme vir Hfst 10 HRW – Rotasie

	9 th Edition	8 th Edition
Problems	6, 28, 33, 40, 45, 53, 59, 80, 82	4, 30, 33, 42, 47, 51, 61, 86, 68

6	4	(a) $\omega_2 = 4.0 \text{ rad/s}$. (b) $\omega_4 = 28 \text{ rad/s}$. (c) 12 rad/s (d) $\alpha_2 = 6.0 \text{ rad/s}^2$ (e) $\alpha_4 = 18 \text{ rad/s}^2$. We note that our answer for α_{avg} does turn out to be the arithmetic average of α_2 and α_4 but point out that this will not always be the case.	
28	30	$t = 13 \text{ s}$	$t = 16 \text{ s}$
33	33	$I = 12.3 \text{ kg}\cdot\text{m}^2$	
40	42	(a) $I = 8.352 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. (b) Comparing to the formula (e) in Table 10-2 (which gives roughly $I = 0.08333 \text{ ML}^2$), we find our answer to part (a) is 0.22% lower.	
45	47	$\tau = (1.30 \text{ m})(4.20 \text{ N}) \sin 75^\circ - (2.15 \text{ m})(4.90 \text{ N}) \sin 60^\circ = -3.85 \text{ N}\cdot\text{m}$.	
53	51	$F_2 = 0.140 \text{ N}$.	
59	61	$P = \tau\omega \Rightarrow \tau = \frac{74600 \text{ W}}{188.5 \text{ rad/s}} = 396 \text{ N}\cdot\text{m}$	
80	68	$\frac{1}{2}(2160)(50)(38)^2(0.052)^2 \leq W \leq \frac{1}{2}(2160)(100)(38)^2(0.052)^2$ $2 \times 10^5 \text{ J} \leq W \leq 4 \times 10^5 \text{ J}.$	
82	86	(a) $\omega_1 = 5.00 \text{ rad/s}$. (b) $\alpha = 1.67 \text{ rad/s}^2$. (c) $\theta_0 = 2.50 \text{ rad}$.	(a) $\omega_1 = 5.00 \text{ rad/s}$. (b) $\alpha = 1.67 \text{ rad/s}^2$. (c) $\theta_0 = 2.50 \text{ rad}$.

Additional Problems for Ch 11 HRW – Angular Momentum
 Addisionele Probleme vir Hfst 11 HRW – Hoekmomentum

	9 th Edition	8 th Edition
Problems	25, 41, 43, 45, 53, 61	23, 41, 51, 43, 55, 61

25	23	(a) $\vec{\tau} = [(3.0\text{m})(6.0\text{N}) - (4.0\text{m})(-8.0\text{N})]\hat{k} = (50\text{N}\cdot\text{m})\hat{k}$. (b) $\phi = 90^\circ$.
41	41	(a) $I_1 + I_2 + I_3 + I_4 = \frac{19}{6}mR^2 = 1.6\text{kg}\cdot\text{m}^2$. (b) $L = I_{\text{total}}\omega = 4.0\text{kg}\cdot\text{m}^2/\text{s}$.
43	51	(a) $r = 1.5\text{ m}$ (b) $\omega = 0.93\text{ rad/s}$. (c) $I = 225\text{ kg}\cdot\text{m}^2$. $K = \frac{1}{2}I\omega^2 = 98\text{ J}.$ (d) $\omega_f = 8.4\text{ rad/s}$. (e) $K_f = 8.8\times 10^2\text{ J}$. (f) We account for the large increase in kinetic energy (part (e) minus part (c)) by noting that the skaters do a great deal of work (converting their internal energy into mechanical energy) as they pull themselves closer — “fighting” what appears to them to be large “centrifugal forces” trying to keep them apart.
45	43	(a) $\omega_f = 3.6\text{ rev/s}$. (b) $\frac{K_f}{K_i} = 3.0$ (c) The man did work in decreasing the rotational inertia by pulling the bricks closer to his body. This energy came from the man’s store of internal energy.
53	55	$I = I_{\text{rod}} + mr^2$ Angular momentum conservation $v = 1.3\times 10^3\text{ m/s}$.
61	61	We make the unconventional choice of <i>clockwise</i> sense as positive, so that the angular velocities in this problem are positive. $\omega = 1.5\text{ rad/s}$.